

Graphs of Functions

Know the six shapes by heart, draw smooth curves from a table, and solve equations by reading where two graphs cross.



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THE TOPIC ON ONE MAP

Graphs of Functions

01 Draw a curve from a table

$$y = x^2 - 4x + 3$$

$$y = x^3 + 2x + 3$$

$$y = \frac{6}{x}$$

02 Solve equations with a graph

read the roots

draw a suitable straight line

03 Recognise the shape

match six graphs

04 Sketch a quadratic

turning point

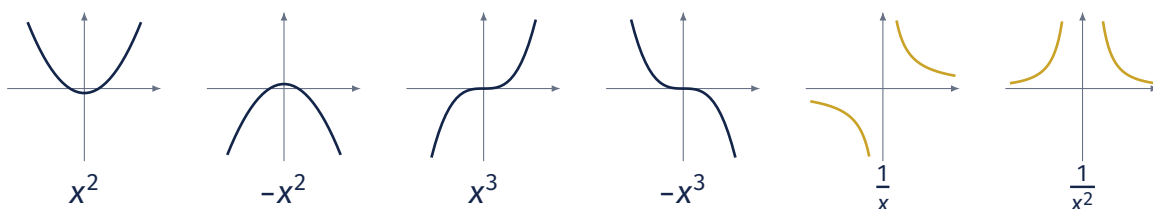
intercepts in terms of a

05 Trigonometric graphs

$\sin x$, $\cos x$, $\tan x$

$$y = a \cos bx + c$$

THE SIX SHAPES Recognise them before you work anything out:



01 Draw a Curve from a Table

Complete the table (calculator: TABLE mode), plot every point, then join with one **smooth** curve — never straight lines between points.

WHEN YOU SEE → YOU DO

“complete the table of values”

→ substitute each x ; brackets round negatives

“draw the graph for $-2 \leq x \leq 4$ ”

→ plot all points, one smooth curve, full range

a quadratic

→ symmetric U or n; the lowest point may be between two table values

$$y = \frac{6}{x}$$

→ two separate branches, never crosses the axes

“write down one thing that is wrong”

→ straight segments / wrong point / pointed bottom

Table (TABLE mode)

Plot every point

One smooth curve

WORKED MODEL · JANUARY 2022 P1HR

Complete the table for $y = x^2 - 4x + 3$, $x = -2$ to 4, and draw the graph.

1 y : 15, 8, 3, 0, -1, 0, 3

substitute

2 plot; smooth U with its lowest point at (2, -1)

draw

ANSWER a smooth U-curve through all 7 points

× **TRAP** Joining the points with a ruler, or a flat bottom. A curve is smooth and curved everywhere.

IDEA 01

Practise: Draw a Curve from a Table

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Table

CH 3 • TOPIC 5 3 marks

Complete a table for $y = x^2 - 2x - 3$ for $x = -2$ to 4 and describe the turning point.

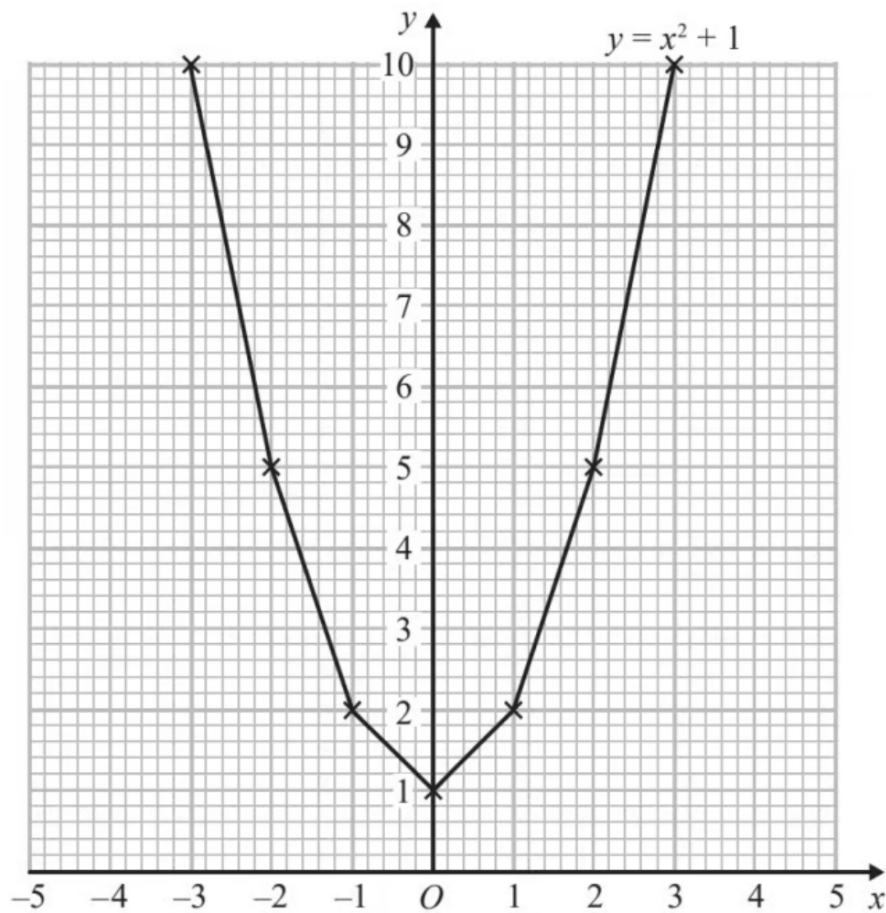
SELECTED PRACTICE

Q2 PRACTICE Practice · Easy

CH 3 · TOPIC 5 1 mark

- 1 Brogan needs to draw the graph of $y = x^2 + 1$

Here is her graph.



Write down one thing that is wrong with Brogan's graph.

(1 mark)

Q3**PRACTICE**

Practice · Easy

CH 3 · TOPIC 5

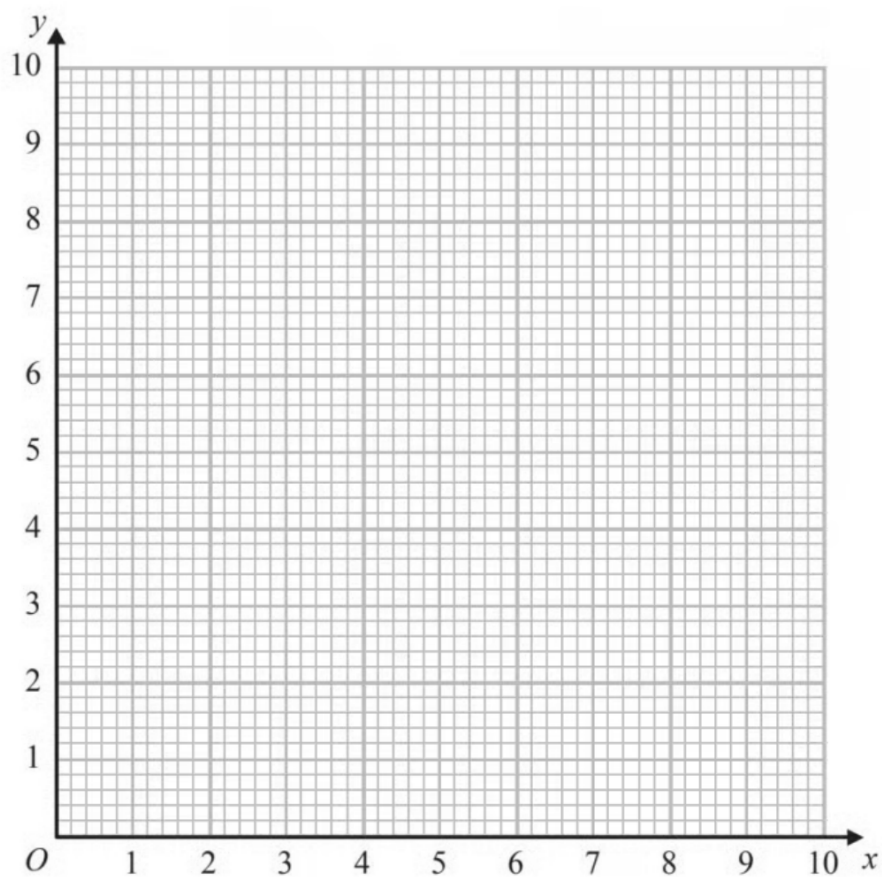
4 marks

5 (a) Complete the table of values for $y = \frac{4}{x}$

x	0.5	1	2	4	5	8
y		4	2			

(2 marks)

(b) On the grid, draw the graph of $y = \frac{4}{x}$ for $0.5 \leq x \leq 8$



(2 marks)

Q4**PRACTICE**

Practice · Medium

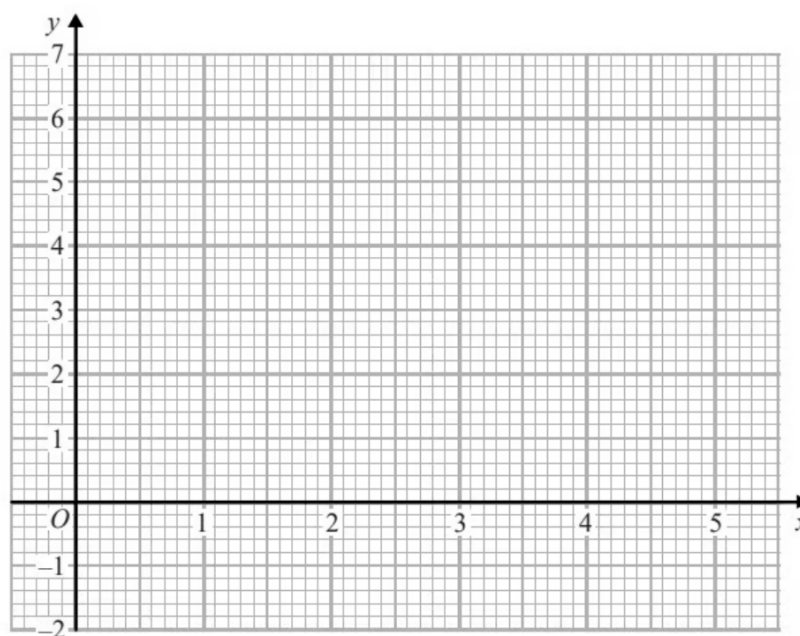
CH 3 · TOPIC 5 6 marks

6 (a) Complete the table of values for $y = x^2 - 5x + 6$

x	0	1	2	3	4	5
y	6		0	0	2	

(1 mark)

(b) On the grid, draw the graph of $y = x^2 - 5x + 6$ for $0 \leq x \leq 5$



(2 marks)

(c) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation

$$x^2 - 5x = x - 7$$

(3 marks)

Q5

PRACTICE

Practice · Hard

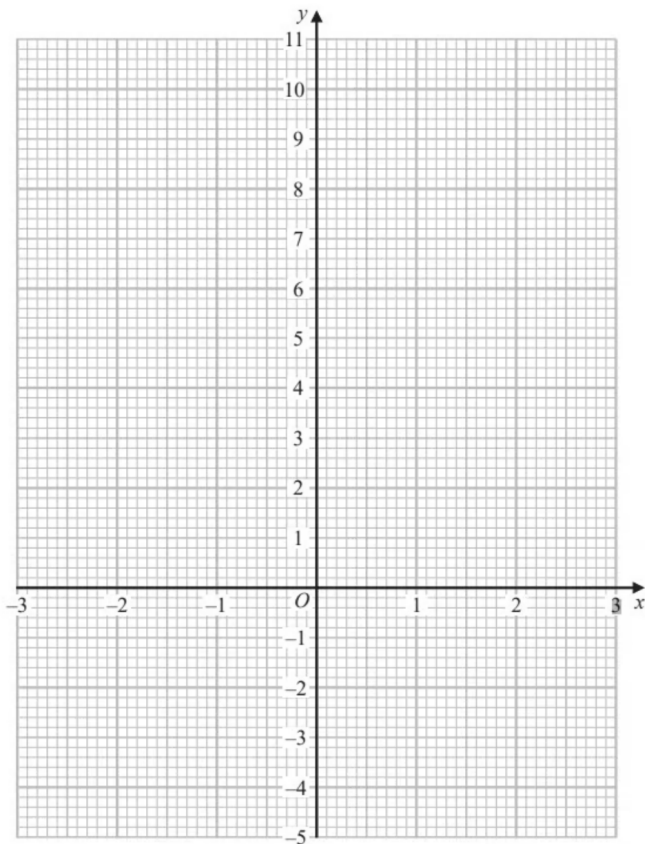
CH 3 · TOPIC 56 marks

7 (a) Complete the table of values for $y = \frac{1}{2}x^3 - 2x + 3$

x	-3	-2	-1	0	1	2	3
y	-4.5			3		3	

(2 marks)

(b) On the grid, draw the graph of $y = \frac{1}{2}x^3 - 2x + 3$ for $-3 \leq x \leq 3$



(2 marks)

(c) By drawing a suitable straight line on the grid, find an estimate for the solution of the equation $\frac{1}{2}x^3 - x + 4 = 0$

$x = \dots\dots\dots$

(2 marks)

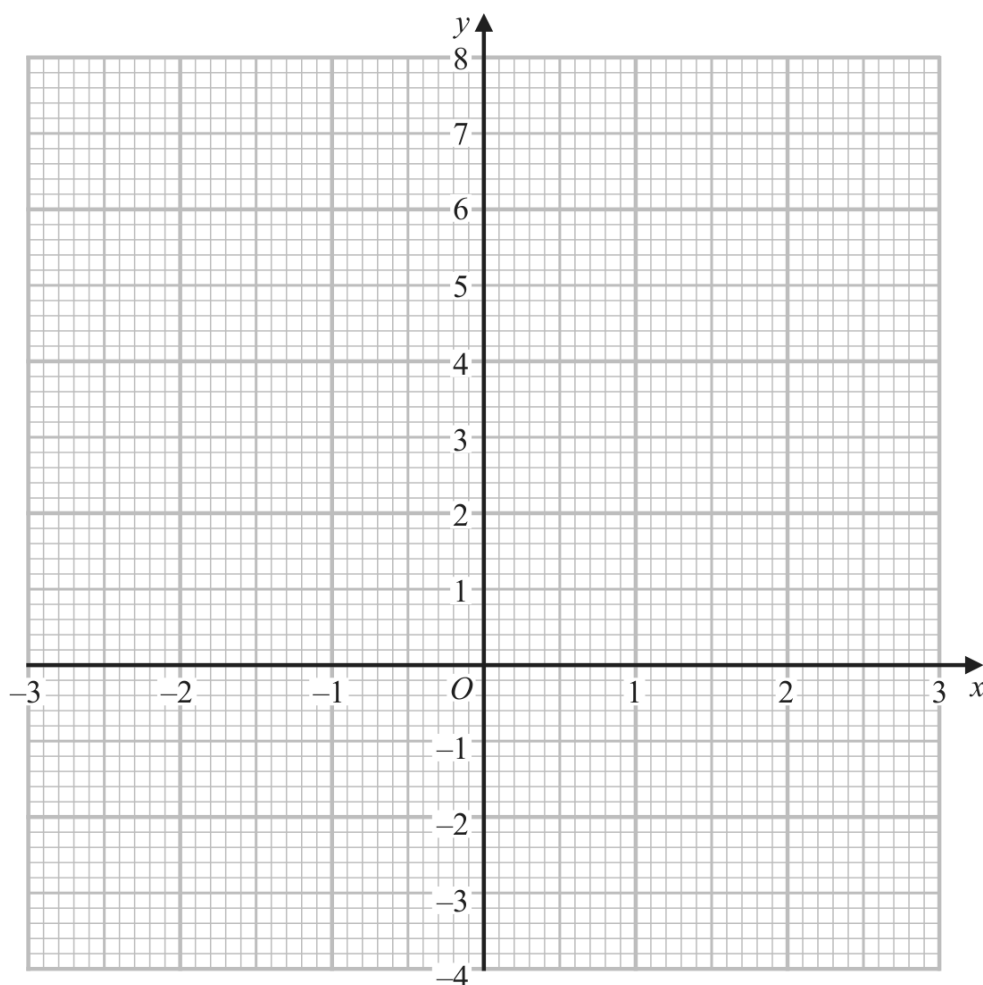
PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2HR · Q15

CH 3 · TOPIC 5 4 marks

15 (a) Complete the table of values for $y = x^2 - \frac{x}{2} - 3$

x	-3	-2	-1	0	1	2	3
y	7.5				-2.5		4.5

(b) On the grid, draw the graph of $y = x^2 - \frac{x}{2} - 3$ for values of x from -3 to 3

(Total for Question 15 is 4 marks)

Q7

PAST PAPER

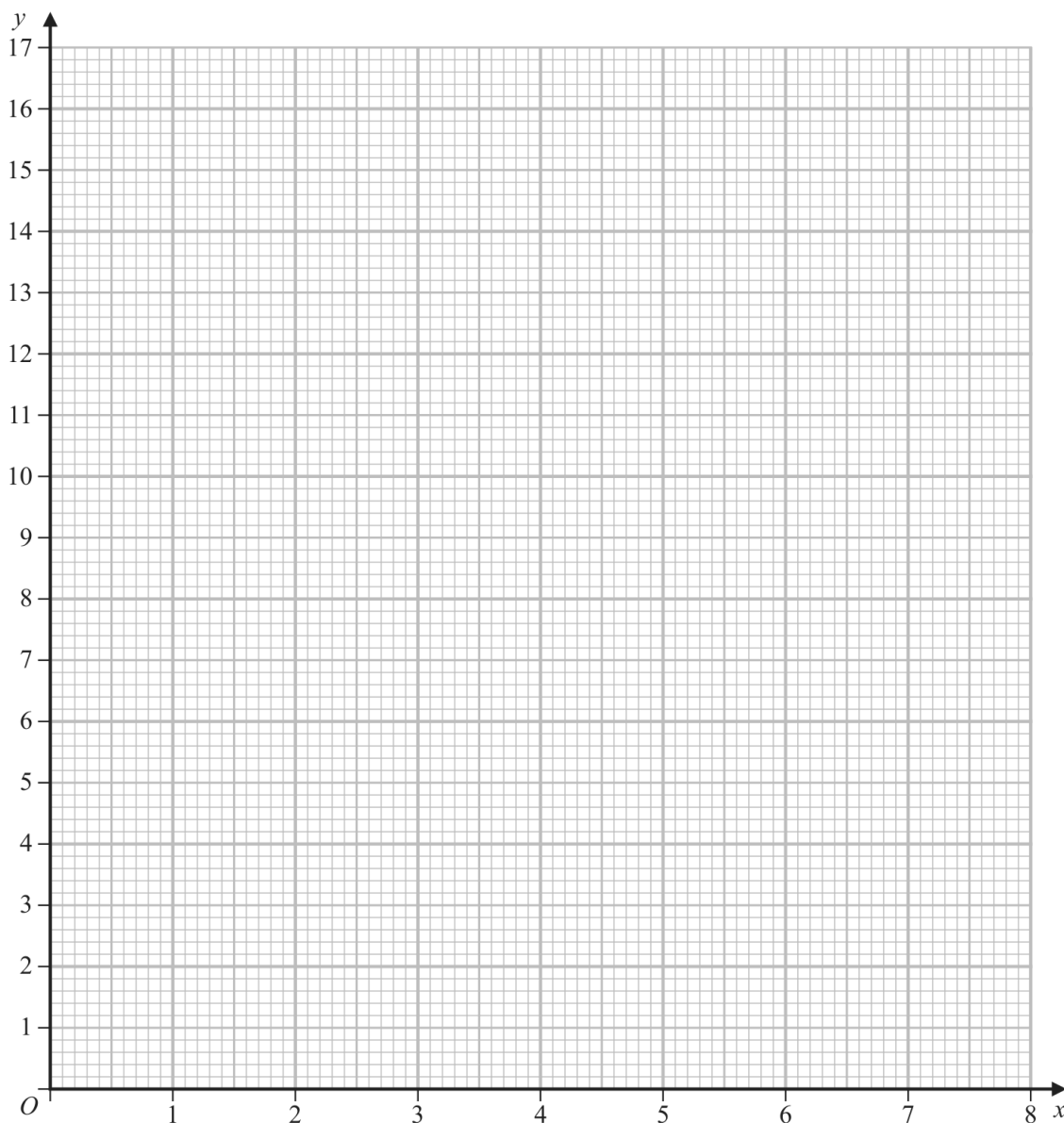
Edexcel 4MA1 • May 2021 • Paper 2H • Q15

CH 3 • TOPIC 5 4 marks

15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leq x \leq 8$



(Total for Question 15 is 4 marks)

Q8**PAST PAPER**

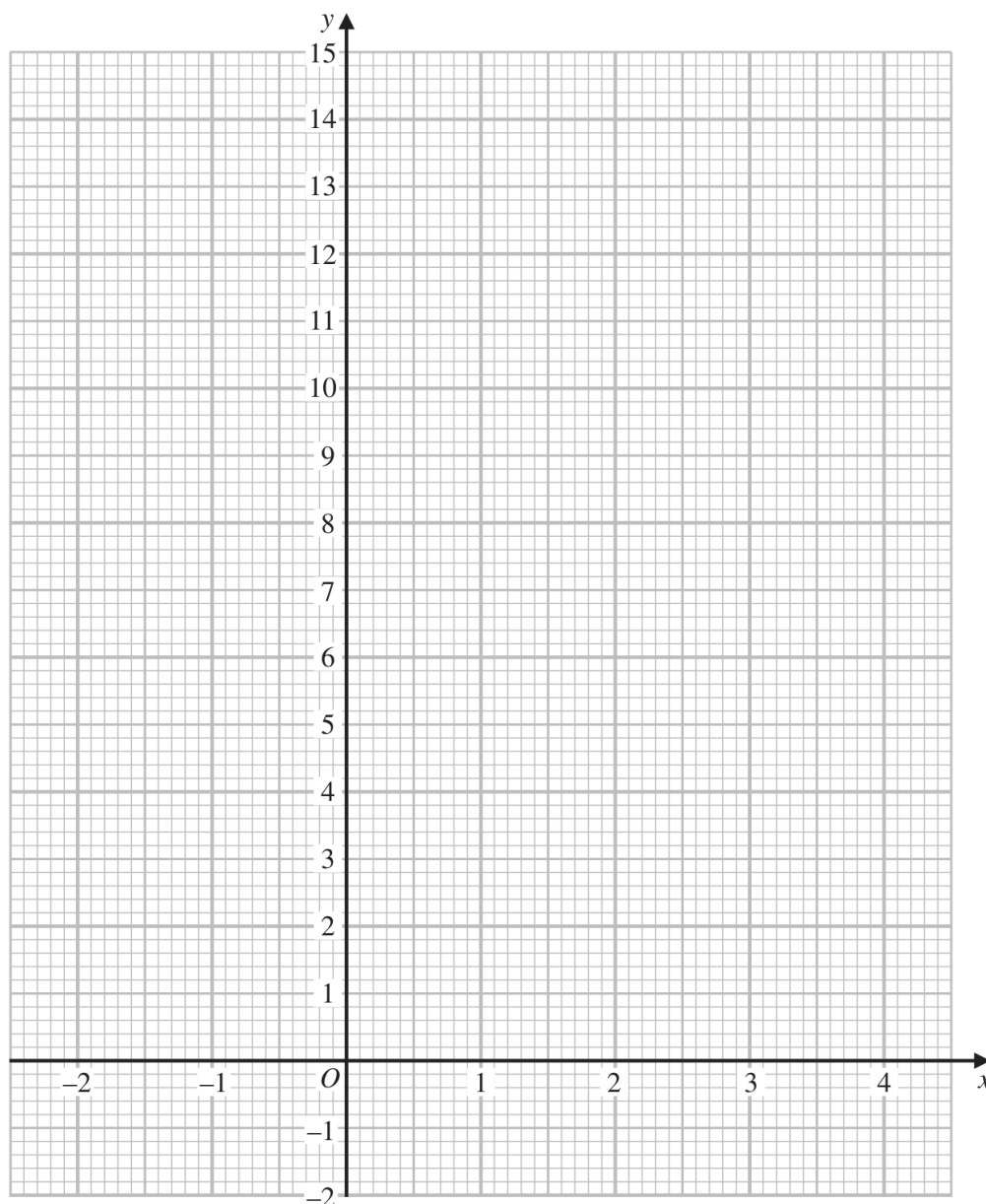
Edexcel 4MA1 · January 2022 · Paper 1HR · Q4

CH 3 · TOPIC 5 4 marks

- 4 (a) Complete the table of values for
- $y = x^2 - 4x + 3$

x	-2	-1	0	1	2	3	4
y		8	3			0	

- (b) On the grid, draw the graph of
- $y = x^2 - 4x + 3$
- for values of
- x
- from -2 to 4



(Total for Question 4 is 4 marks)

Q9**PAST PAPER**

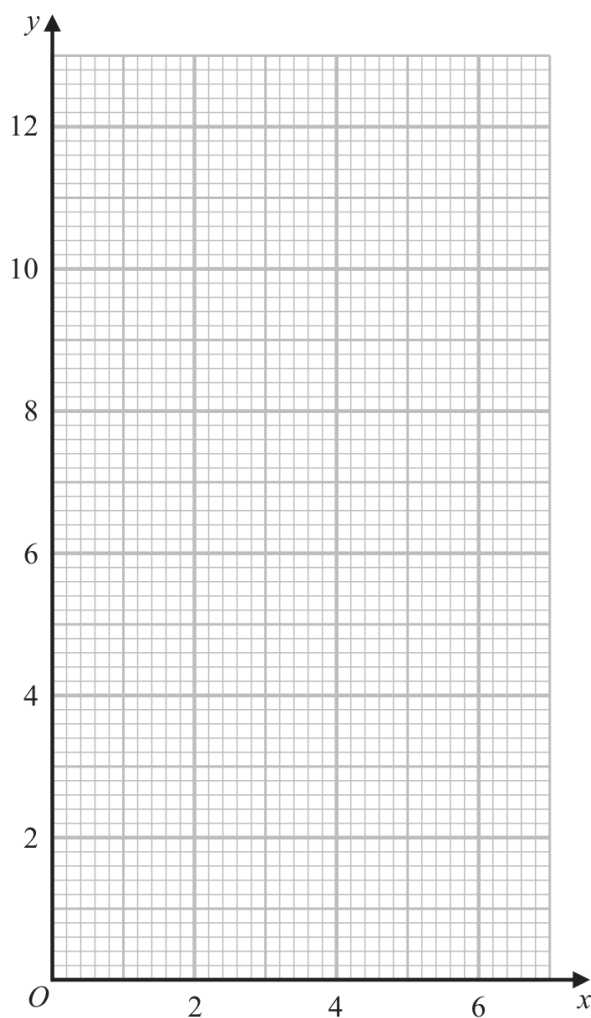
Edexcel 4MA1 · January 2022 · Paper 2HR · Q11

CH 3 · TOPIC 5 4 marks

- 11 (a) Complete the table of values for $y = \frac{6}{x}$

x	0.5	1	2	3	4	5	6
y		6		2			1

- (b) On the grid, draw the graph of $y = \frac{6}{x}$ for $0.5 \leq x \leq 6$



(Total for Question 11 is 4 marks)

Q10**PAST PAPER**

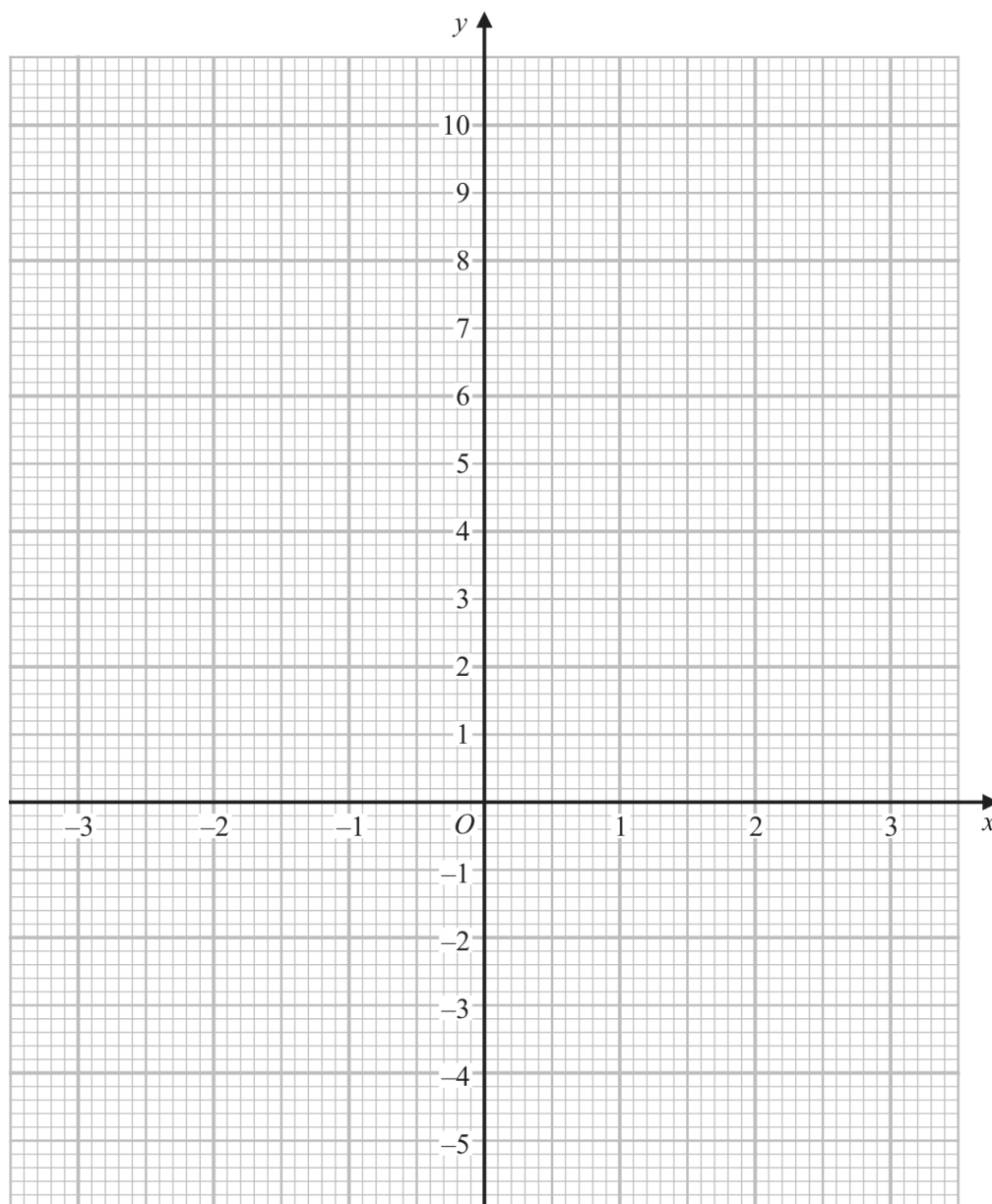
Edexcel 4MA1 • June 2023 • Paper 2H • Q4

CH 3 • TOPIC 5 4 marks

- 4 (a) Complete the table of values for
- $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

- (b) On the grid below, draw the graph of
- $y = x^2 - x - 4$
- for values of
- x
- from -3 to 3



(Total for Question 4 is 4 marks)

Q11**PAST PAPER**

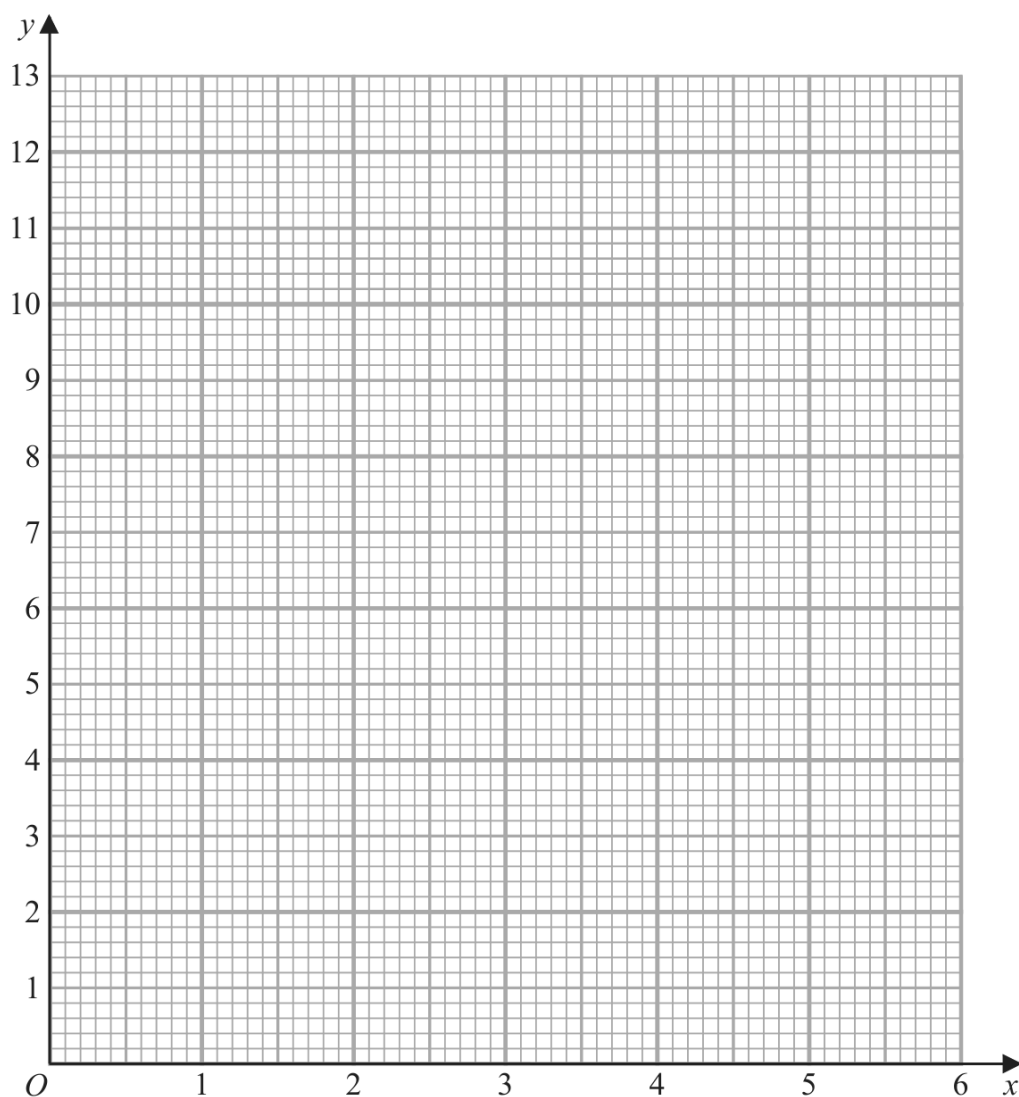
Edexcel 4MA1 · November 2024 · Paper 1H · Q11

CH 3 · TOPIC 5 4 marks

- 11 (a) Complete the table of values for $y = 2\left(x + \frac{1}{x}\right)$

x	0.5	1	2	3	4	5	6
y			5	6.7			12.3

- (b) On the grid, draw the graph of $y = 2\left(x + \frac{1}{x}\right)$ for $0.5 \leq x \leq 6$



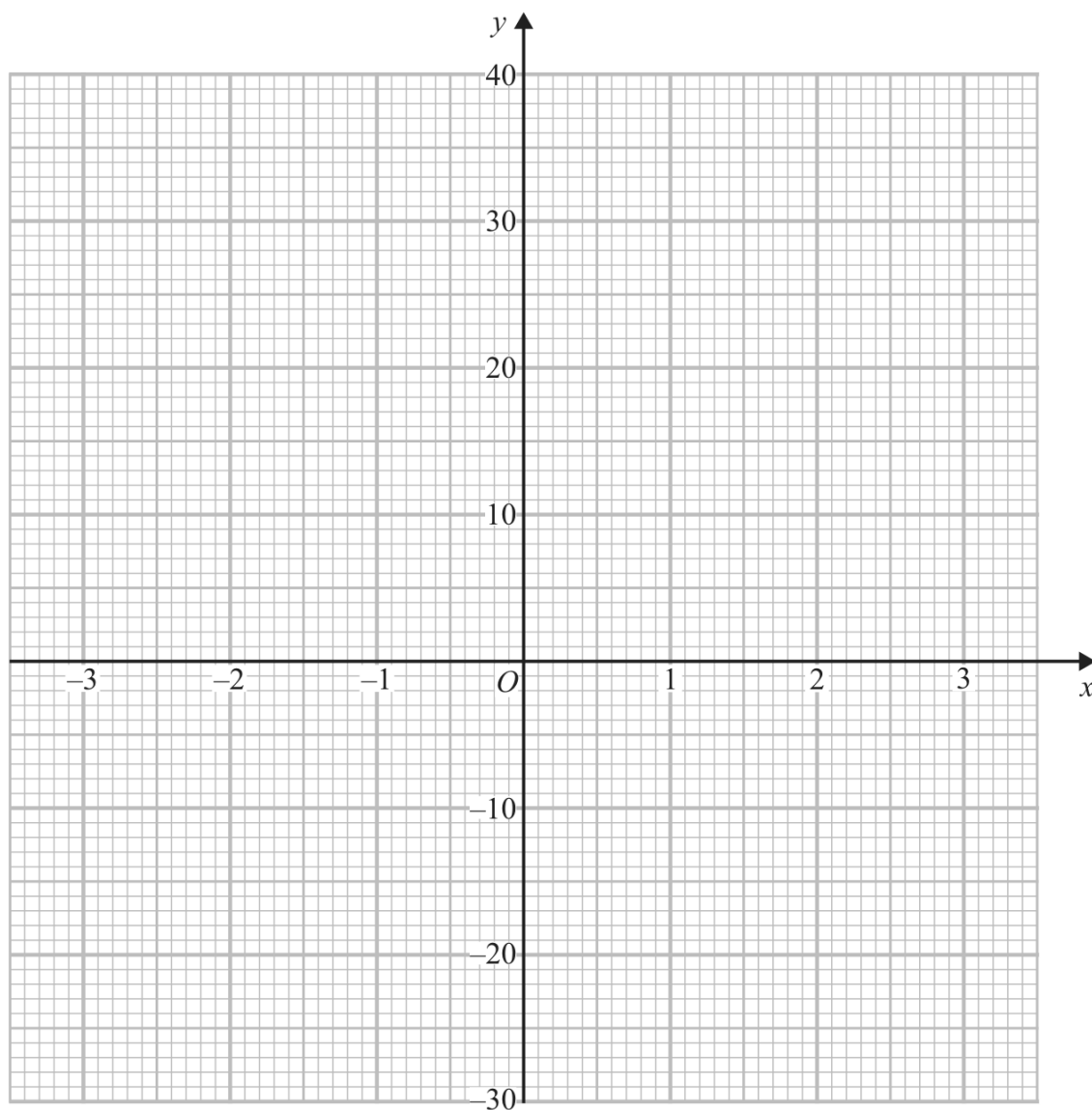
(Total for Question 11 is 4 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q10

CH 3 • TOPIC 5 3 marks10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leq x \leq 3$ **(Total for Question 10 is 3 marks)**

02 Solve Equations with a Graph

The solutions of $f(x) = 0$ are where the curve crosses the x -axis. For any other equation, rearrange it into “curve = line”, draw the line, and read the x values where they cross.

WHEN YOU SEE → YOU DO

“use the graph to solve $2x^2 - 4x - 1 = 0$ ” → read where the curve crosses $y = 0$

“solve $x^2 - x - 6 = -2$ ” → draw $y = -2$, read the crossings

“by drawing a suitable straight line” → (given equation) - (graph) = the line

“correct to one decimal place” → read carefully from the small squares

Graph equation

Subtract the new equation

Line to draw

Read x

WORKED MODEL · JANUARY 2022 P2H

The graph of $y = 2x^2 - 4x - 1$ is drawn. Use it to solve $x^2 - x - 1 = 0$.

1 $\times 2$: $2x^2 - 2x - 2 = 0$

match the x^2

2 $2x^2 - 4x - 1 = -2x + 1$

graph = line

ANSWER draw $y = 1 - 2x$: $x \approx -0.6, 1.6$

IDEA 02

Practise: Solve Equations with a Graph

My questions first, then selected practice, then real past papers.

Q13 DR ESAM Suitable line

CH 3 • TOPIC 5 2 marks

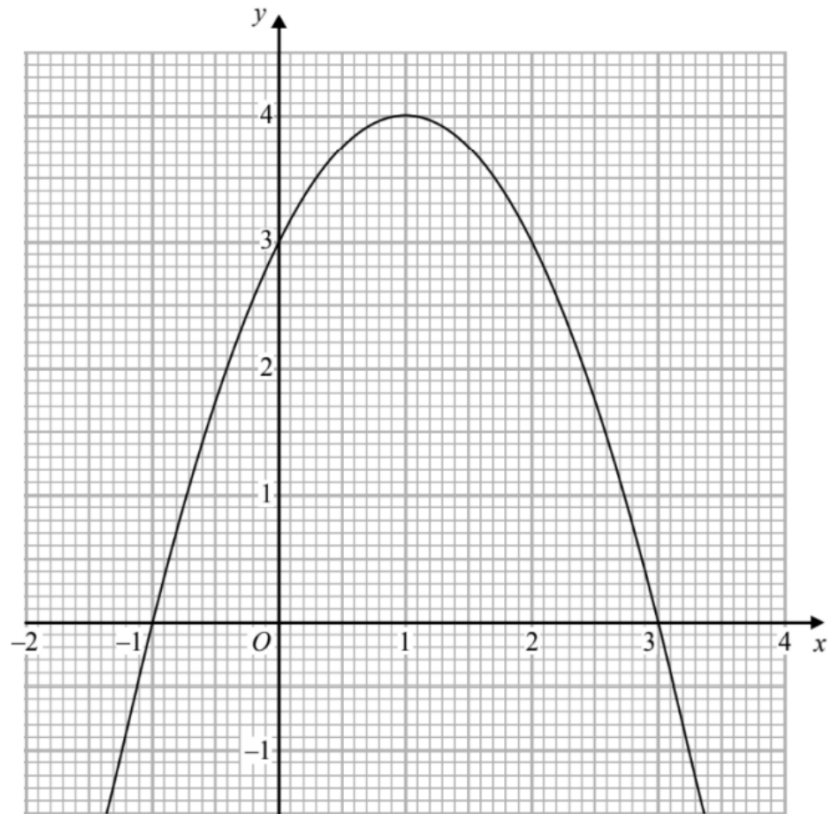
The graph of $y = x^2 - 3x$ is drawn. Which line would you draw to solve $x^2 - 4x + 1 = 0$?

SELECTED PRACTICE

Q14 PRACTICE Practice · Easy

CH 3 · TOPIC 5 3 marks

6 (a) The graph of $y = f(x)$ is drawn on the grid.



Write down the coordinates of the turning point of the graph.

(1 mark)

(b) Write down the roots of $f(x) = 2$

(1 mark)

(c) Write down the value of $f(0.5)$

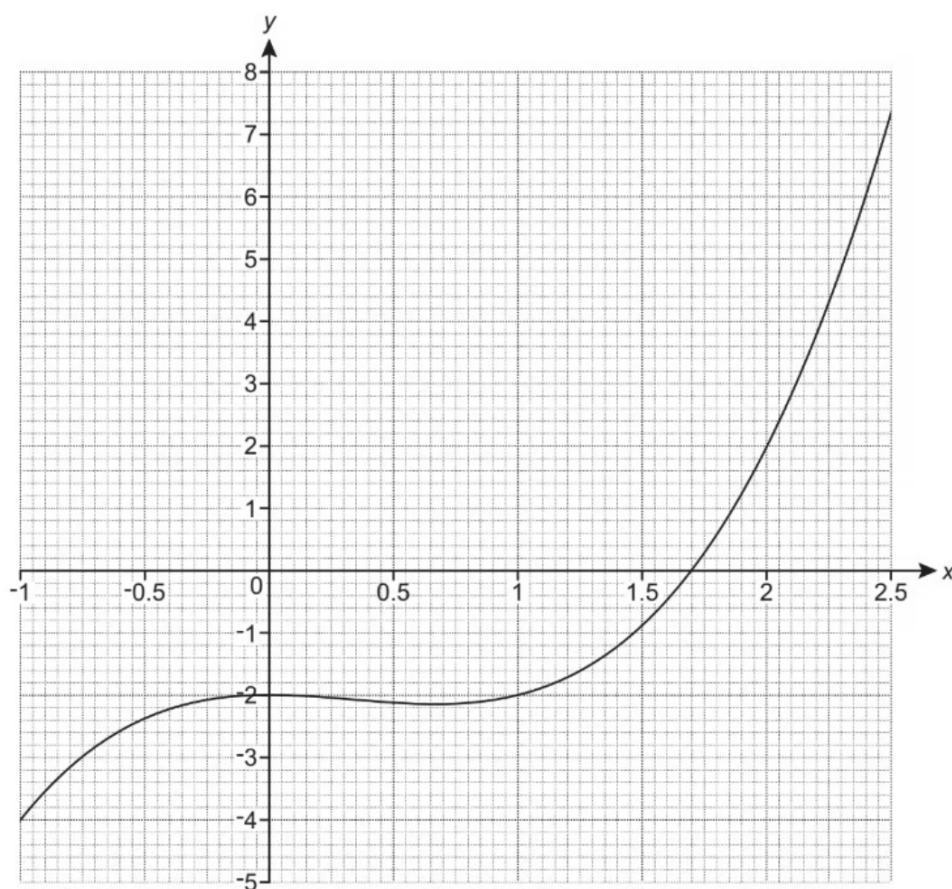
(1 mark)

Q15**PRACTICE**

Practice · Easy

CH 3 · TOPIC 5 1 mark

14 The graph of $y = x^3 - x^2 - 2$ is drawn on the grid.



Use the graph to solve $x^3 - x^2 - 2 = 0$.

Give your answer correct to 1 decimal place.

$x = \dots\dots\dots$

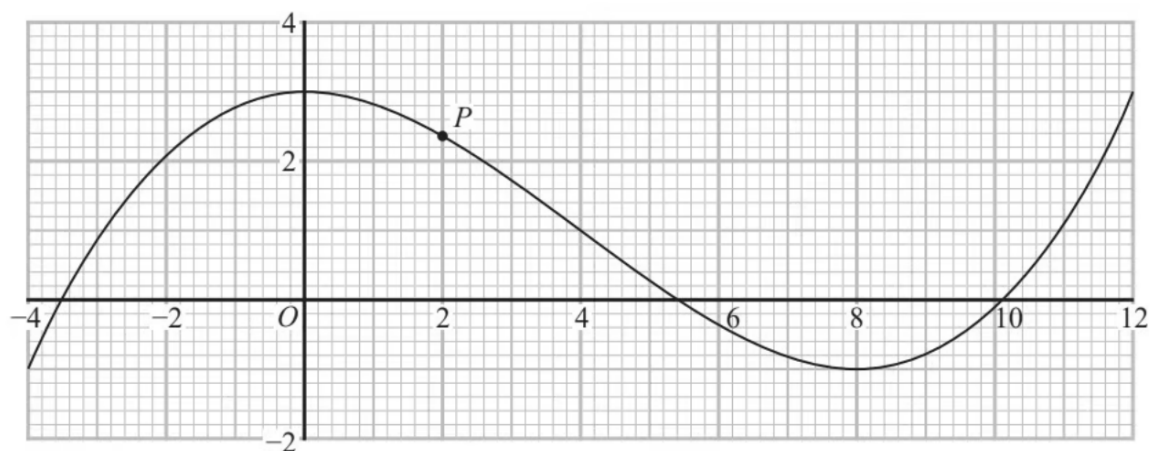
(1 mark)

Q16**PRACTICE**

Practice · Medium

CH 3 · TOPIC 5 2 marks

- 8 The diagram shows the graph of $y = f(x)$ for $-4 \leq x \leq 12$



Note: Point P in the graph is from the previous question. It is not required for this question.

The equation $f(x) = k$ has exactly two different solutions for $-4 \leq x \leq 12$

Use the graph to find the two possible values of k .

(2 marks)

Q17

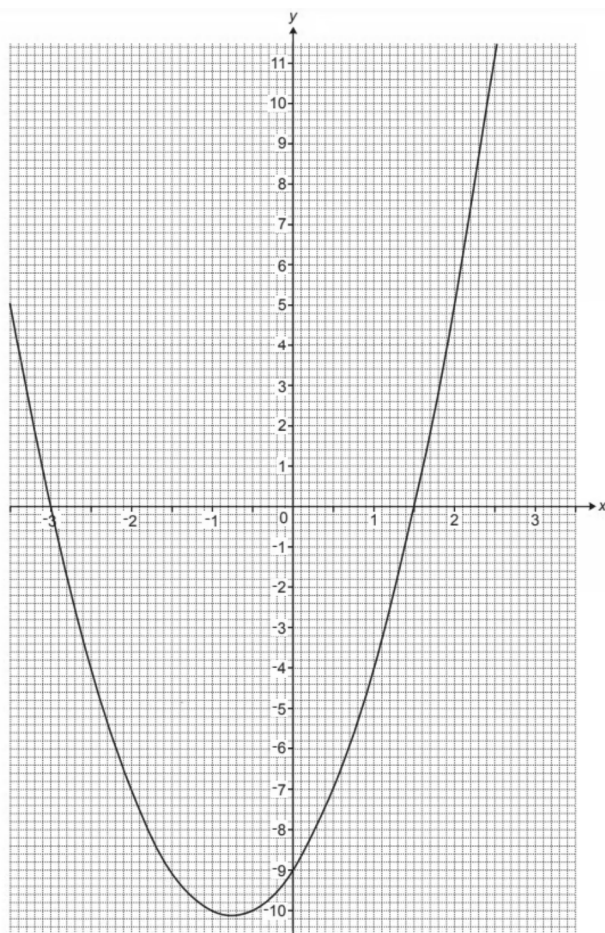
PRACTICE

Practice · Hard

CH 3 · TOPIC 5

4 marks

13 (a) The graph of $y = 2x^2 + 3x - 9$ is drawn below.



Use the graph to solve $2x^2 + 3x - 9 = 0$.

$x = \dots\dots\dots$ or

$x = \dots\dots\dots$

(2 marks)

(b) The equation $2x^2 + x - 4 = 0$ can be solved by finding the intersection of the graph of $y = 2x^2 + 3x - 9$ and the line $y = ax + b$.

(i) Find the value of a and the value of b .

$a = \dots\dots\dots$

$b = \dots\dots\dots[2]$

(ii) Hence use the graph to solve the equation $2x^2 + x - 4 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots[3]$

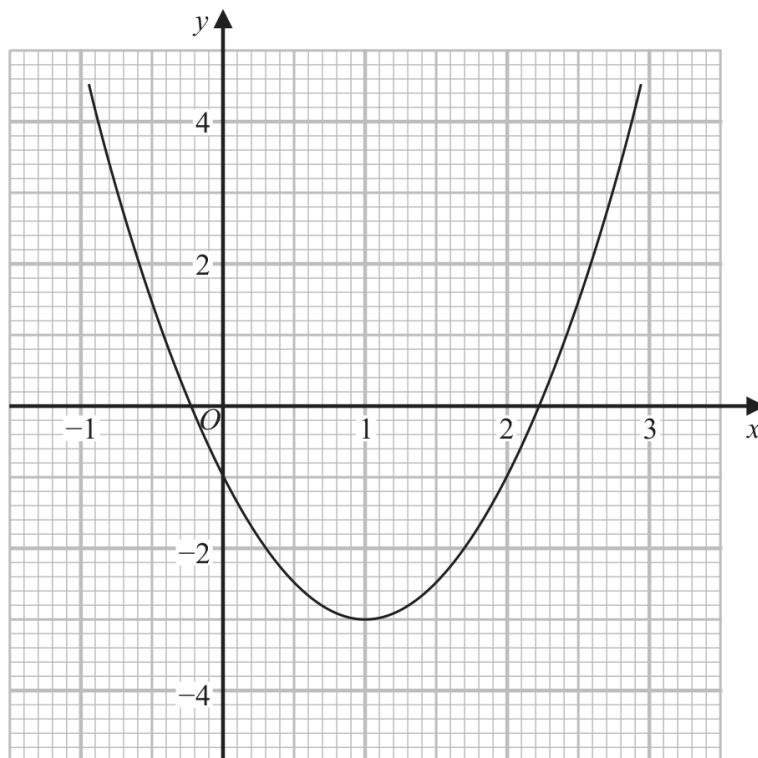
(2 marks)

PAST PAPERS

Q18 PAST PAPER Edexcel 4MA1 · January 2022 · Paper 2H · Q21

CH 3 · TOPIC 5 5 marks

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

.....

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....

(Total for Question 21 is 5 marks)

Q19**PAST PAPER**

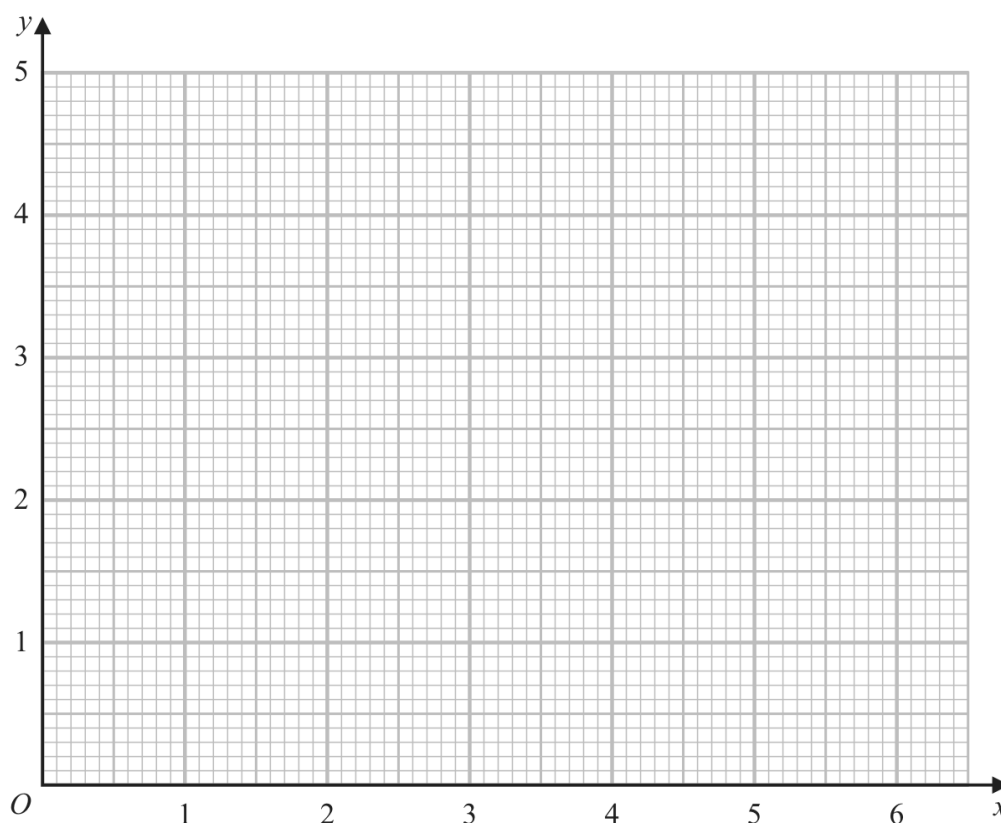
Edexcel 4MA1 • November 2025 • Paper 2H • Q12

CH 3 • TOPIC 5 5 marks

12 (a) Complete the table of values for $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$

x	0.5	1	2	3	4	5	6
y	4.25			2.17	2.5	2.9	3.33

(b) Draw the graph of $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$ for $0.5 \leq x \leq 6$



(c) By drawing a suitable line on the grid, find estimates for the solutions of

the equation $x + \frac{4}{x} = 6$

Give your answers correct to one decimal place.

(Total for Question 12 is 5 marks)

03 Recognise the Shape

Look at the highest power, then the sign in front. A power on the bottom ($\frac{1}{x}$, $\frac{1}{x^2}$) means the curve never touches the axes.

WHEN YOU SEE → YOU DO

x^2 / $-x^2$	→ U / n
x^3 / $-x^3$	→ rises left to right / falls left to right
$\frac{k}{x}$, $k > 0$	→ top-right and bottom-left
$-\frac{k}{x}$	→ top-left and bottom-right
$\frac{k}{x^2}$	→ both branches above the x-axis
$\sin x$, $\cos x$	→ waves between -1 and 1; $\tan x$ has asymptotes

ZONE 1

$y = +\frac{k}{x}$: **opposite quarters**

top-right & bottom-left

ZONE 2

$y = -\frac{k}{x}$: **the other two**

top-left & bottom-right

ZONE 3

$y = \frac{k}{x^2}$: **both above**

symmetric in the y-axis

IDEA 03

Practise: Recognise the Shape

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q20****PRACTICE**

Practice · Easy

CH 3 · TOPIC 5 1 mark

- 12 Circle the point that is on the graph of $y = \frac{1}{x}$

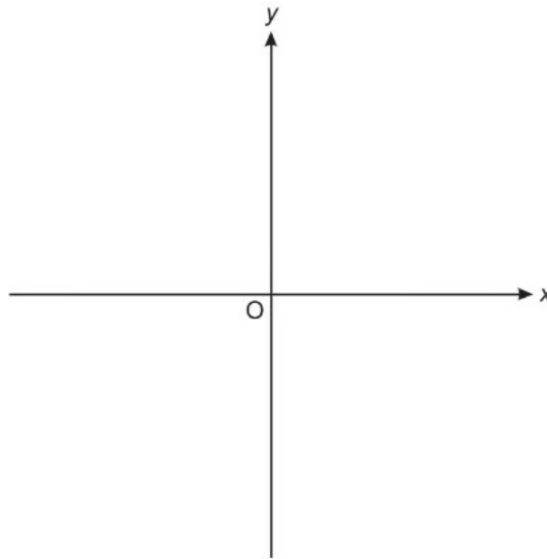
$(-1, 1)$	$(0.3, 3)$	$(0.8, 0.2)$	$(2.5, 0.4)$
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(1 mark)

Q21 **PRACTICE** Practice · Easy

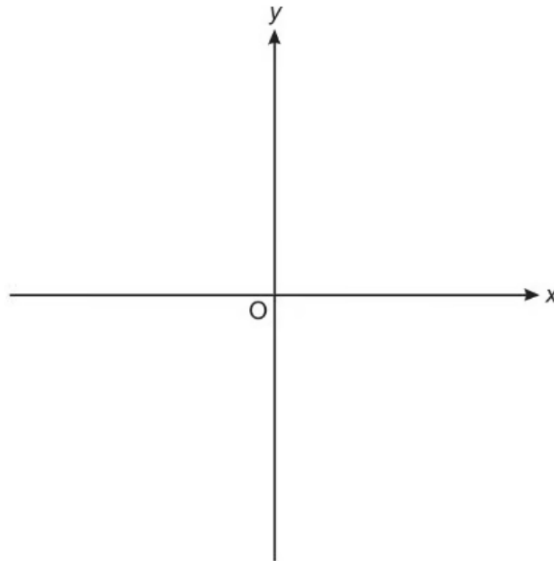
CH 3 · TOPIC 5 2 marks

- 16 (i) Sketch a graph on the axes that shows that y is directly proportional to x .



[2]

- (ii) Sketch a graph on the axes that shows $y = x^3$



[2]

(2 marks)

Q22**PRACTICE**

Practice · Medium

CH 3 · TOPIC 5 1 mark

10 Circle the point that lies on the curve $y = x^2 - 4x + 1$

$(-1, 4)$	$(-1, -4)$	$(-1, -2)$	$(-1, 6)$
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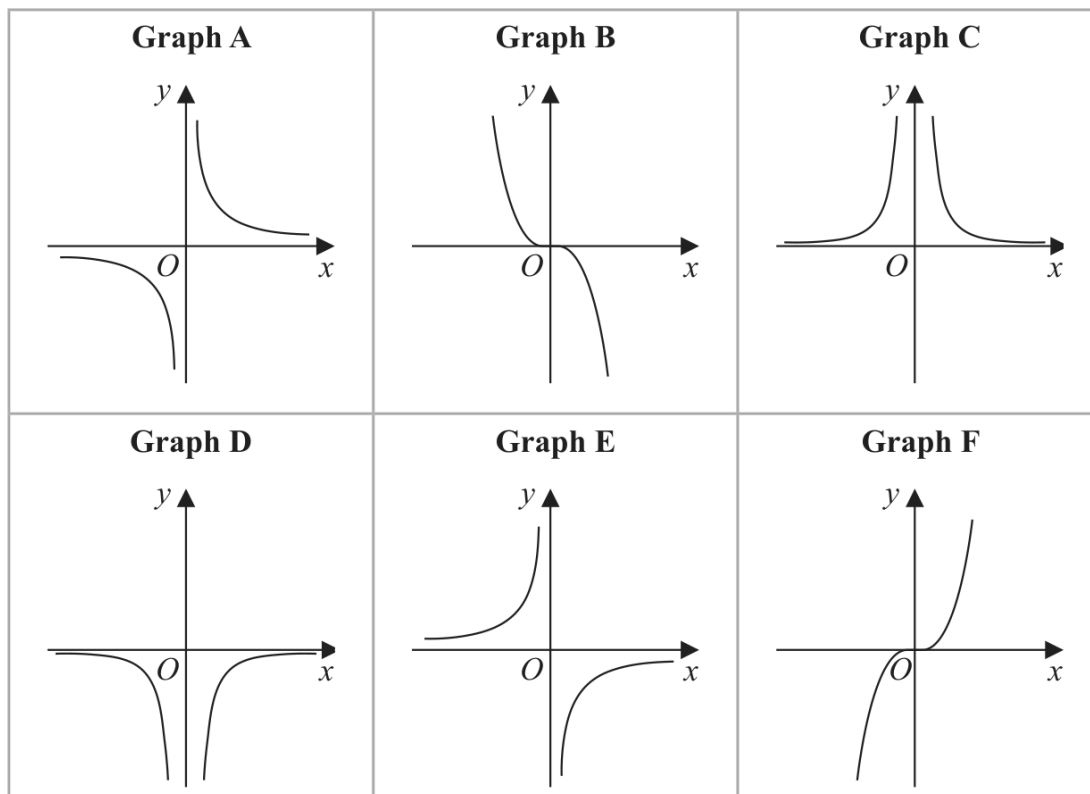
(1 mark)

PAST PAPERS

Q23 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 2H • Q15

CH 3 • TOPIC 5 3 marks

15 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \frac{2}{x^2}$	
$y = -\frac{1}{2}x^3$	
$y = -\frac{5}{x}$	

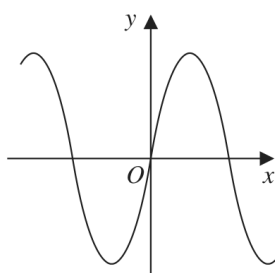
(Total for Question 15 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 2HR • Q16

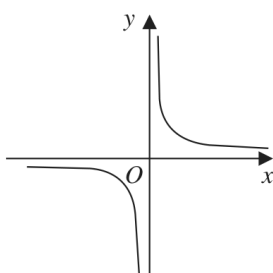
CH 3 • TOPIC 5 **3 marks**

16 Here are nine graphs.

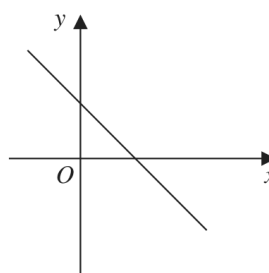
Graph A



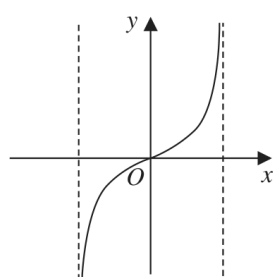
Graph B



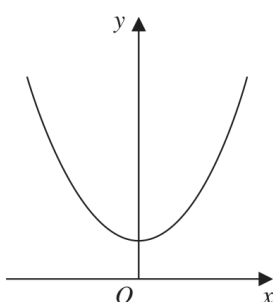
Graph C



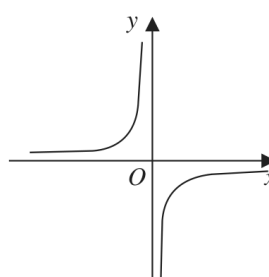
Graph D



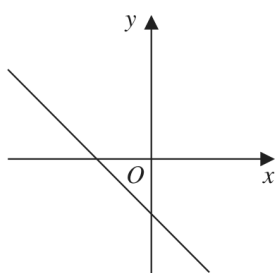
Graph E



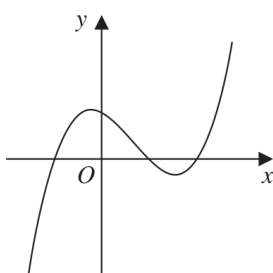
Graph F



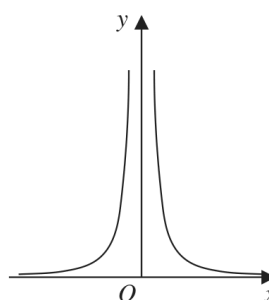
Graph G



Graph H



Graph I



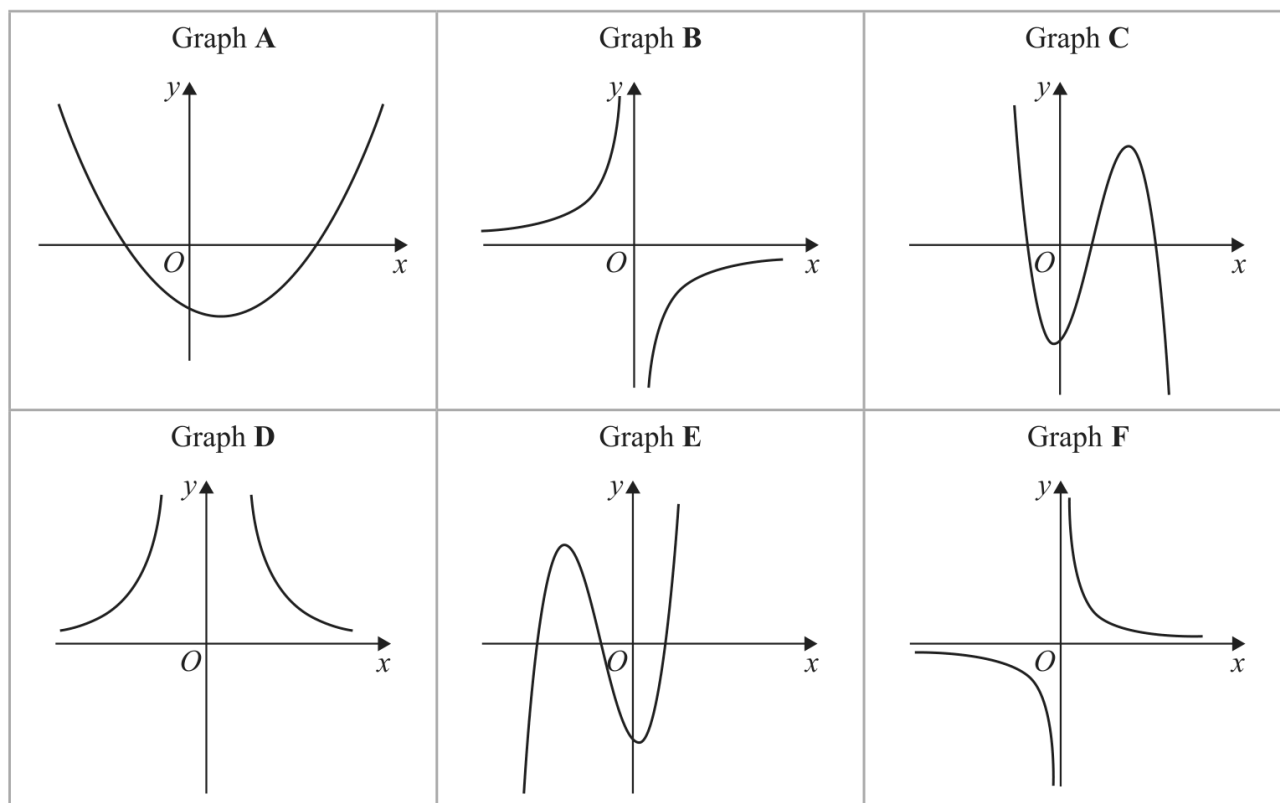
Complete the table below with the letter of the graph that could represent each given equation.
Write each answer on the dotted line.

Equation	Graph
$y = -2x + 3$	
$y = -\frac{1}{x}$	
$y = \tan x^\circ$	
$y = (x + 1)(x - 1)(x - 2)$	

(Total for Question 16 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q12

CH 3 • TOPIC 5 3 marks**12** Here are six graphs.

Write down the letter of the graph of

(a) $y = \frac{10}{x^2}$

.....

(b) $y = x - 3 + 3x^2 - x^3$

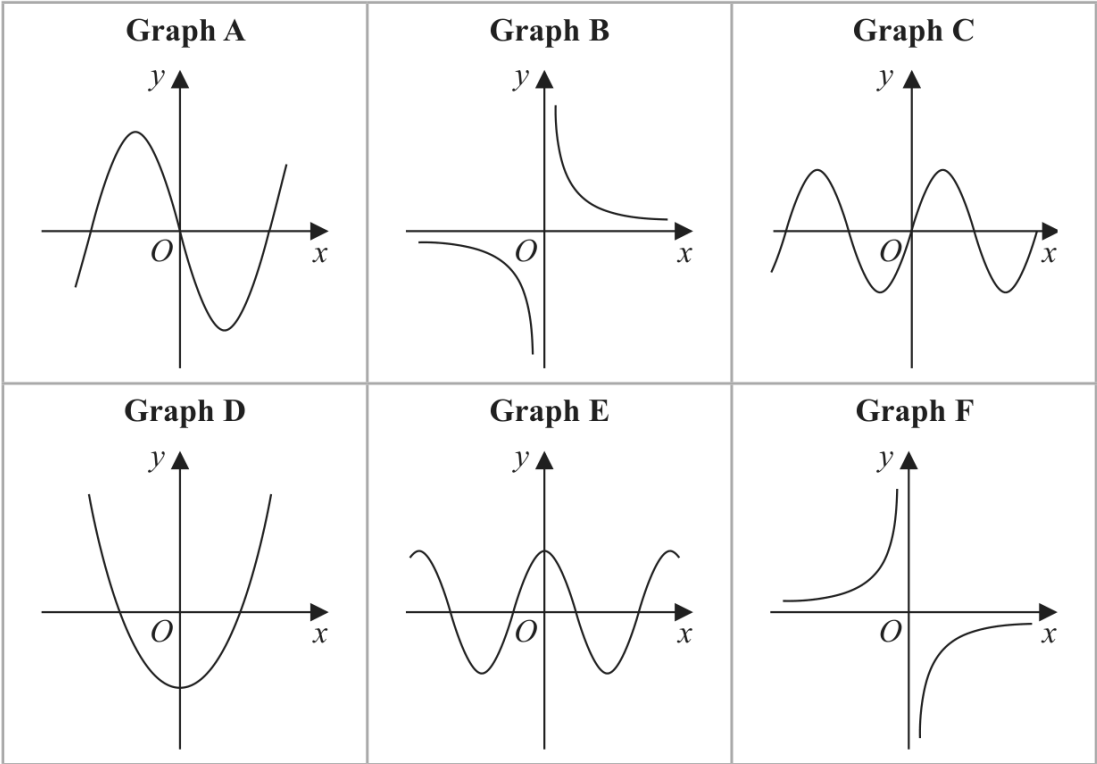
.....

(c) $y = -\frac{3}{x}$

.....

(Total for Question 12 is 3 marks)

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	
$y = -\frac{3}{x}$	
$y = 4x^3 - 5x$	

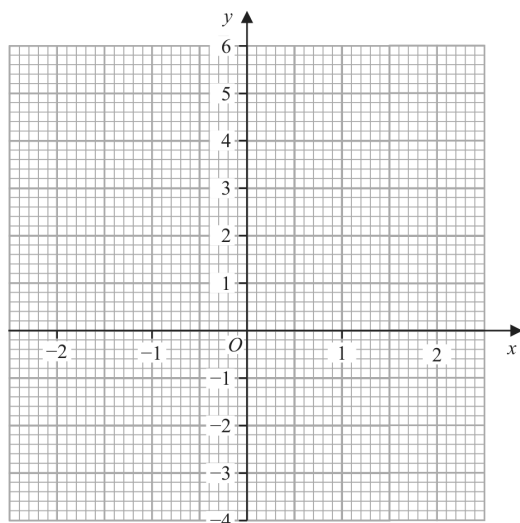
(Total for Question 18 is 3 marks)

Q27**PAST PAPER**

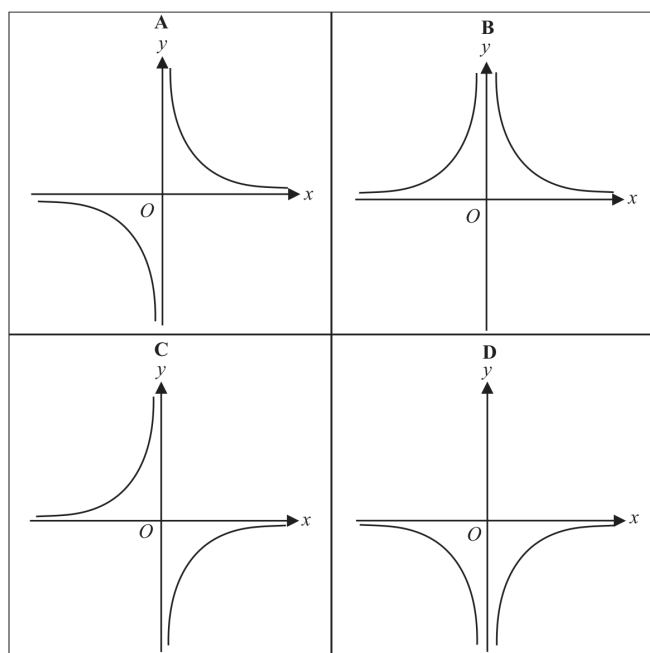
Edexcel 4MA1 • November 2023 • Paper 1H • Q12

CH 3 • TOPIC 5 5 marks12 (a) Complete the table of values for $y = x^3 - 3x + 1$

x	-2	-1	0	1	2
y				-1	

(b) On the grid, draw the graph of $y = x^3 - 3x + 1$ for values of x from -2 to 2

Here are four graphs.

(c) Write down the letter of the graph that could have the equation $y = -\frac{1}{x^2}$

(Total for Question 12 is 5 marks)

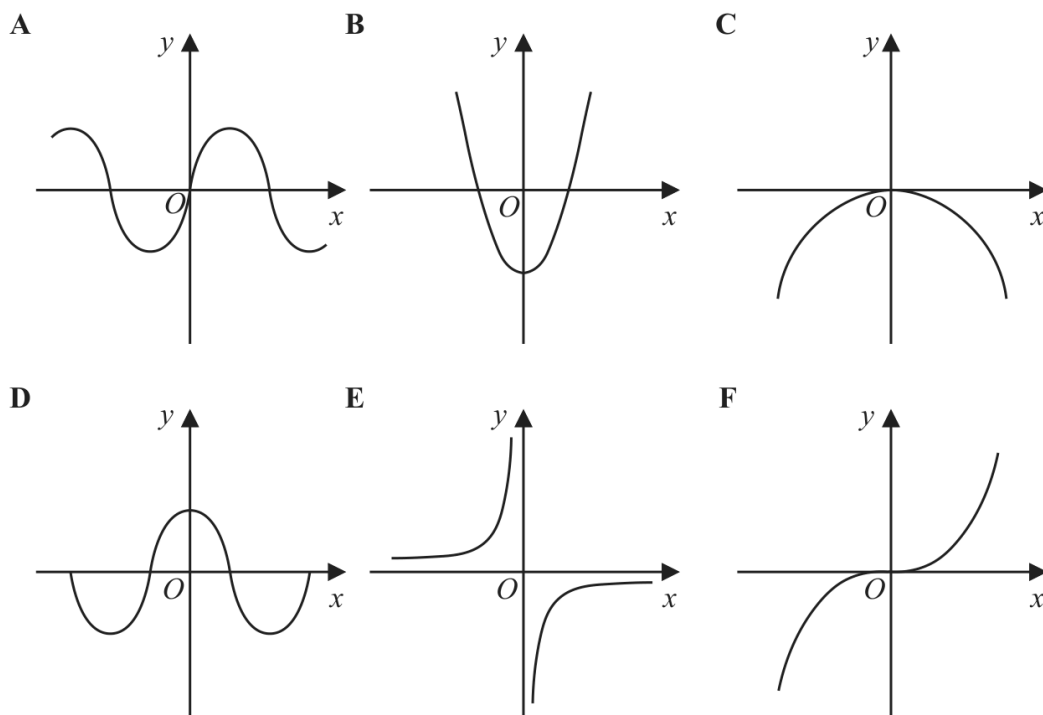
Q28

PAST PAPER

Edexcel 4MA1 • June 2024 • Paper 2HR • Q16

CH 3 • TOPIC 5 2 marks

16 Here are six graphs.



Write down the letter of the graph that could have the equation

(i) $y = -\frac{1}{x}$

.....

(ii) $y = \sin x^\circ$

.....

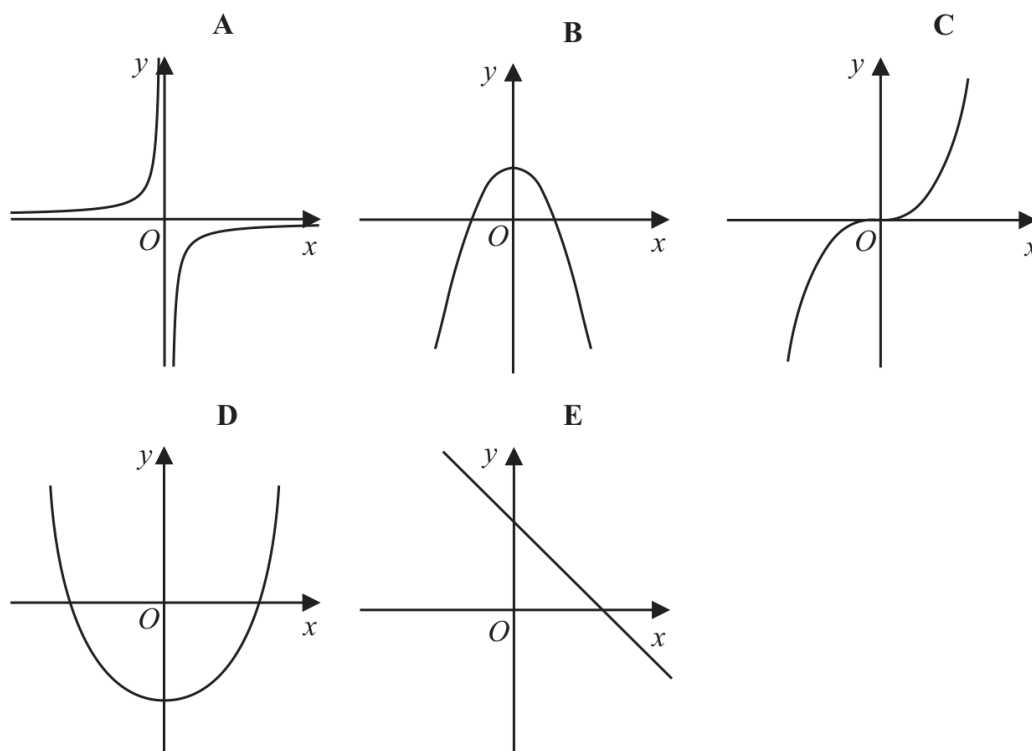
(Total for Question 16 is 2 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q14

CH 3 • TOPIC 5 2 marks

14 Here are five graphs.

(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$

.....

(b) Write down the letter of the graph that could have the equation $y = 2x^3$

.....

(Total for Question 14 is 2 marks)

04 Sketch a Quadratic

Complete the square for the turning point; put $x = 0$ for the y-intercept; put $y = 0$ for the x-intercepts. Then sketch the shape through them.

WHEN YOU SEE → YOU DO

“sketch, show the turning point”

→ complete the square: $(x-p)^2+q \rightarrow (p, q)$

“exact coordinates of intersection with the axes”

→ $x = 0$; and solve $y = 0$ (surds allowed)

“in terms of a ”

→ keep a : roots $1 \pm \frac{\sqrt{a}}{2}$

“maximum point” ($-x^2$)

→ n shape: $a - b(x+c)^2 \rightarrow (-c, a)$

Complete the square

Turning point

y-intercept

x-intercepts

Sketch

WORKED MODEL • JANUARY 2020 P1H

Sketch $y = 4(x - 1)^2 - a$, where $a > 4$.

1 turning point $(1, -a)$

flip inside, keep outside

2 $x = 0$: $y = 4 - a$ (below O , since $a > 4$)

y-intercept

3 $(x - 1)^2 = \frac{a}{4} \Rightarrow x = 1 \pm \frac{\sqrt{a}}{2}$

x-intercepts

ANSWER U-shape through $(0, 4 - a)$, $\left(1 \pm \frac{\sqrt{a}}{2}, 0\right)$

IDEA 04

Practise: Sketch a Quadratic

My questions first, then selected practice, then real past papers.

Q30**DR ESLAM**

Sketch

CH 3 · TOPIC 5

3 marks

Sketch $y = x^2 - 6x + 5$, showing the turning point and where it meets the axes.

SELECTED PRACTICE**Q31****PRACTICE**

Practice · Easy

CH 3 · TOPIC 5

1 mark

13 The equation of a curve is $y = (x - 1)^2 - 6$

Circle the coordinates of the turning point.

$(-1, -6)$	$(1, 6)$	$(-1, 6)$	$(1, -6)$
------------	----------	-----------	-----------

(1 mark)

Q32**PRACTICE**

Practice · Hard

CH 3 · TOPIC 5 1 mark

- 8 The curve **C** has equation $y = 3 - 5(x + 1)^2$
The point **A** is the maximum point on **C**.

Write down the coordinates of **A**.

(1 mark)

Q33**PRACTICE**

Practice · Hard

CH 3 · TOPIC 5 3 marks

10 (a) The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$

The point **A** is the maximum point on **C**.

Write down the coordinates of **A**.

(1 mark)

(b) The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ Find an equation for the curve **S**.

(1 mark)

(c) The curve **C** is transformed to the curve **T**.

The curve **T** has equation $y = 3(x + 2)^2 - 9$

Describe fully the transformation that maps curve **C** onto curve **T**.

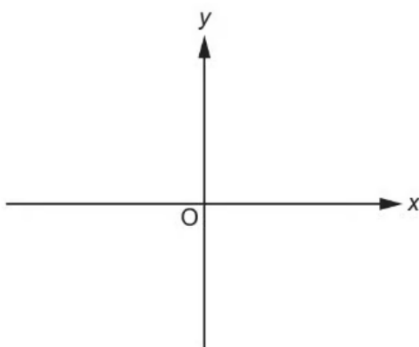
(1 mark)

Q34**PRACTICE**

Practice · Hard

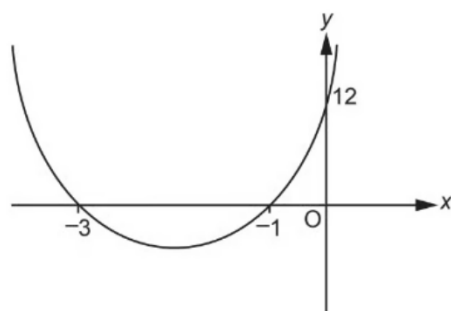
CH 3 · TOPIC 5 8 marks

- 14 (a) Sketch the graph of $y = (x-2)^2 - 3$. Show the coordinates of any turning points.



(3 marks)

- (b) The sketch shows part of a graph which has equation $y = ax^2 + bx + c$.



Not to scale

Find the values of a , b and c .

 $a = \dots\dots\dots$
 $b = \dots\dots\dots$
 $c = \dots\dots\dots$

(5 marks)

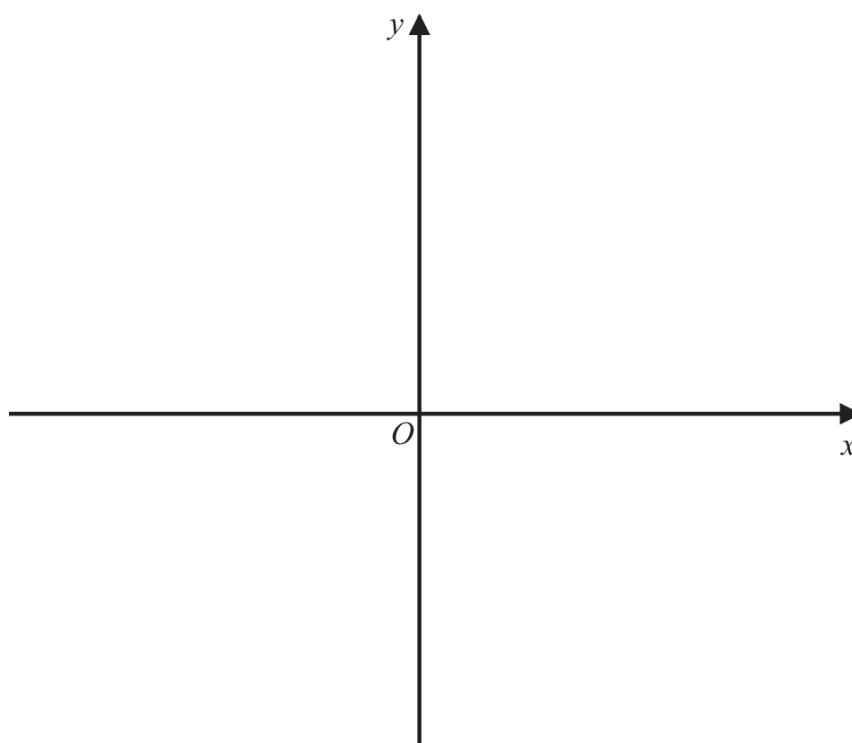
PAST PAPERS

Q35 **PAST PAPER** Edexcel 4MA1 · January 2020 · Paper 1H · Q20**CH 3 · TOPIC 5** **4 marks**

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

Q36**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q23

CH 3 · TOPIC 5 5 marks

23 The curve **C** has equation $y = x^2 - 8x - 9$

The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)

The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k

Show clear algebraic working.

$k =$

(Total for Question 23 is 5 marks)

05 Trigonometric Graphs

$\sin$ and $\cos$ repeat every 360° between -1 and 1 . In $y = a \cos bx + c$: a = height from the middle, b = how many waves in 360° , c = the middle line.

WHEN YOU SEE → YOU DO

“the coordinates of P and Q on $y = \sin x$ ”

→ $P(180, 0), Q(270, -1)$

“sketch $y = \tan x$ ”

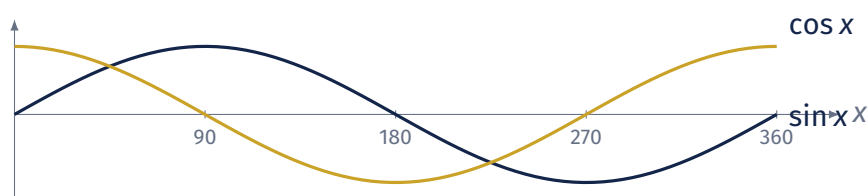
→ through $0, 180, 360$; asymptotes at 90 and 270

“find a, b, c ”

→ $a = \frac{\max - \min}{2}, c = \frac{\max + \min}{2}, b = \frac{360}{\text{period}}$

$\cos(180 - \alpha), \cos(360 + \alpha)$

→ use the symmetry of the graph: $-k, k$



WORKED MODEL · JUNE 2023 P2HR

$y = a \cos bx + c$: max 3 , min -1 , four waves from 0° to 360° .

1 $a = \frac{3 - (-1)}{2} = 2, c = \frac{3 + (-1)}{2} = 1$

height and middle

2 period $90^\circ \rightarrow b = \frac{360}{90}$

number of waves

ANSWER $a = 2, b = 4, c = 1$

IDEA 05

Practise: Trigonometric Graphs

My questions first, then selected practice, then real past papers.

Q37 DR ESLAM Trig

CH 3 • TOPIC 5 2 marks

Write down the coordinates of the minimum point of $y = \cos x$ for $0 \leq x \leq 360$.

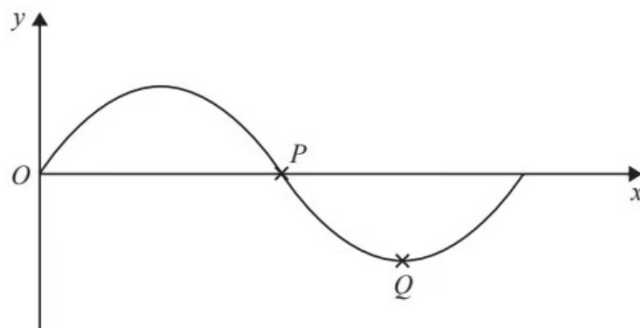
SELECTED PRACTICE

Q38**PRACTICE**

Practice · Easy

CH 3 · TOPIC 5 4 marks

- 10 (a) The diagram shows part of a sketch of the curve $y = \sin x^\circ$



Write down the coordinates of

- (i) the point P

[1]

- (ii) the point Q

[1]

(2 marks)

- (b) Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$

Show the coordinates of any points of intersection with the coordinate axes.

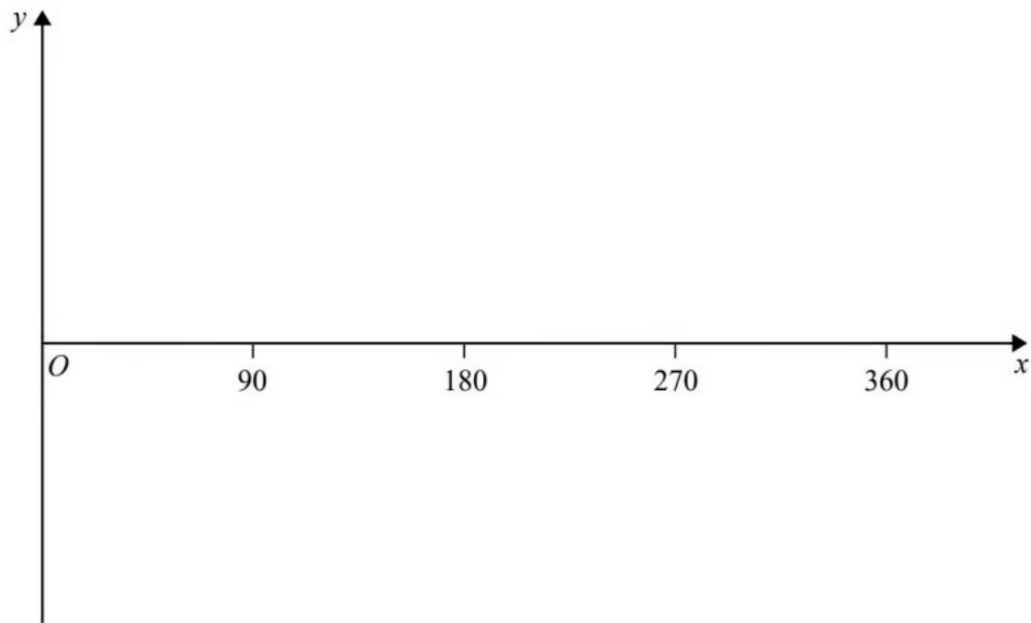
**(2 marks)**

Q39**PRACTICE**

Practice · Medium

CH 3 · TOPIC 5 2 marks

2 Sketch the graph of $y = \cos x^\circ$ for $0 \leq x \leq 360$



(2 marks)

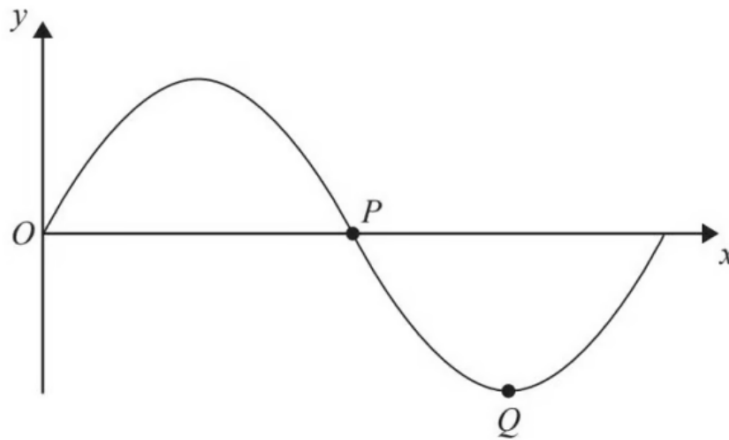
Q40**PRACTICE**

Practice · Medium

CH 3 · TOPIC 5

2 marks

3 (a) The diagram shows part of a sketch of the curve $y = \sin x^\circ$.



Write down the coordinates of the point P .

(1 mark)

(b) Write down the coordinates of the point Q .

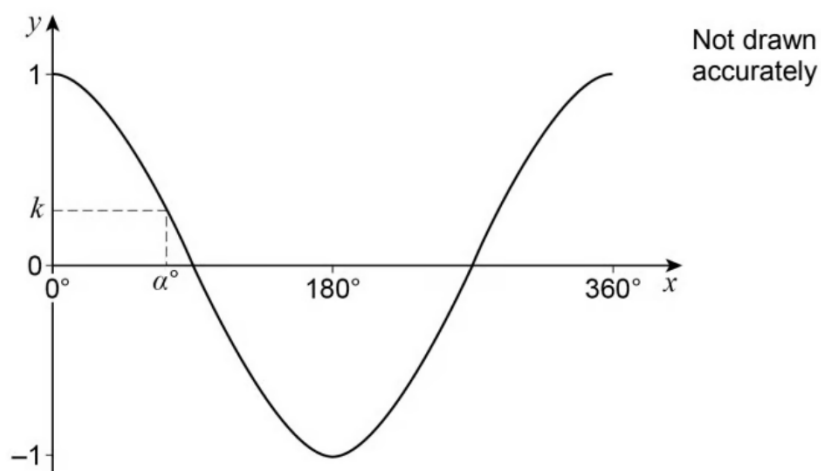
(1 mark)

Q41**PRACTICE**

Practice · Medium

CH 3 · TOPIC 5 2 marks

11 (a) Here is a sketch of $y = \cos x$ for values of x from 0° to 360°



α° is an acute angle.

$$\cos \alpha^\circ = k$$

Circle the value of $\cos (180^\circ - \alpha^\circ)$

$1 - k$	k	$-k$	$-1 - k$
---------	-----	------	----------

(1 mark)

(b) Circle the value of $\cos (360^\circ + \alpha^\circ)$

$k - 1$	$k + 1$	$-k$	k
---------	---------	------	-----

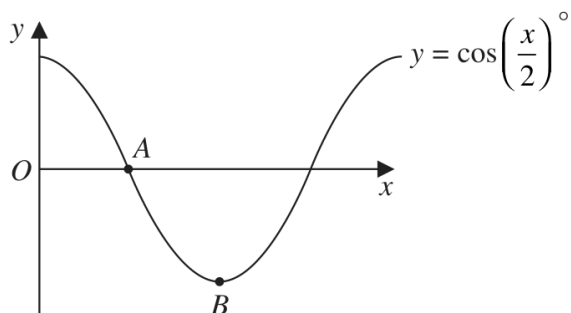
(1 mark)

PAST PAPERS

Q42 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 2H • Q23

CH 3 • TOPIC 5 2 marks

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(i) Find the coordinates of the point A

(.....,)

(ii) Find the coordinates of the point B

(.....,)

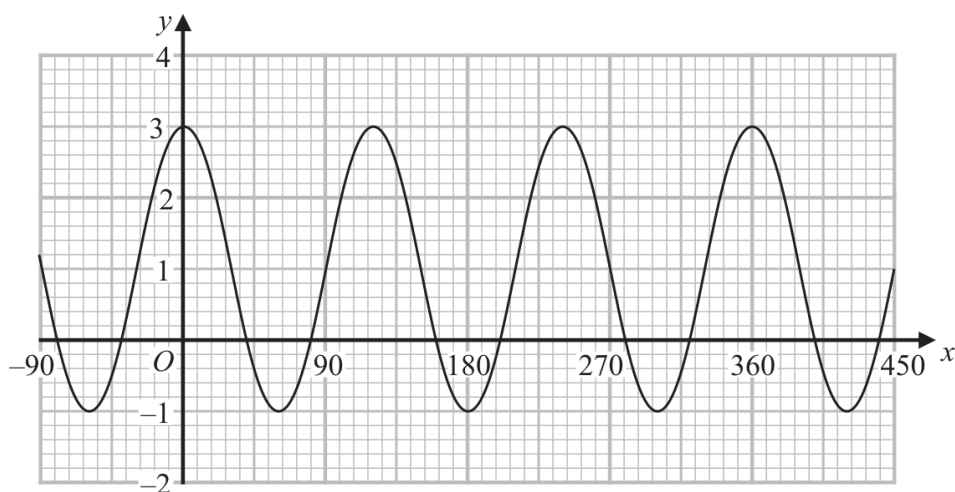
(Total for Question 23 is 2 marks)

Q43**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q26

CH 3 · TOPIC 5 3 marks

26 Here is a sketch of the curve with equation $y = a \cos bx^\circ + c$ where $-90 \leq x \leq 450$



Find the value of a , the value of b and the value of c

$a = \dots\dots\dots$

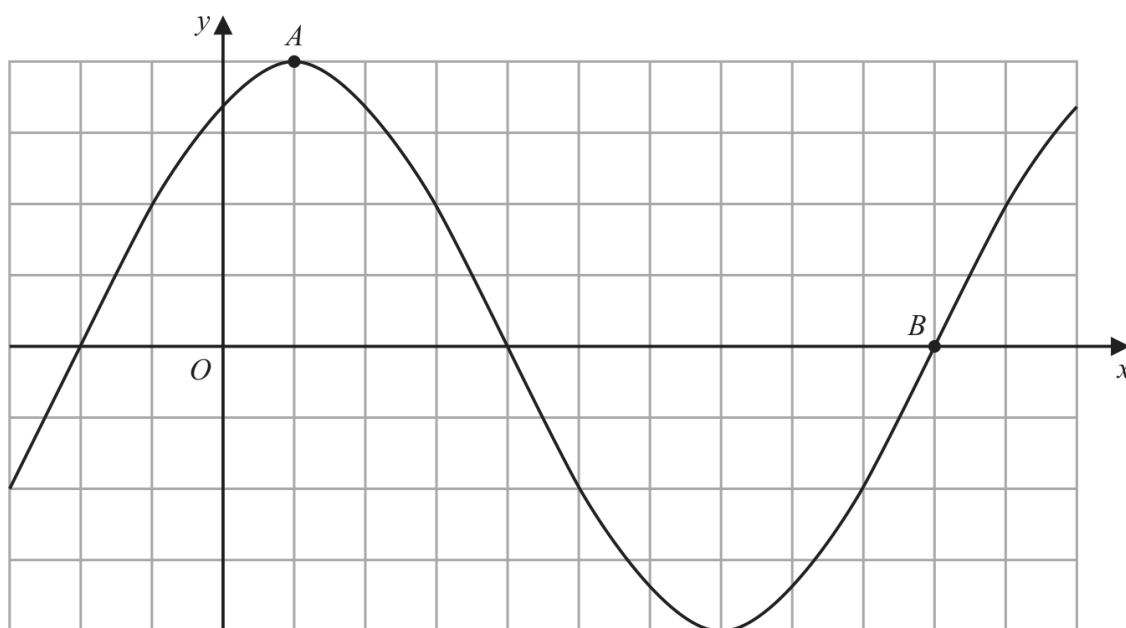
$b = \dots\dots\dots$

$c = \dots\dots\dots$

(Total for Question 26 is 3 marks)

Q44 PAST PAPER Edexcel 4MA1 · June 2024 · Paper 2H · Q25

CH 3 · TOPIC 5 2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$ (i) Find the coordinates of the point A

(.....,)

(ii) Find the coordinates of the point B

(.....,)

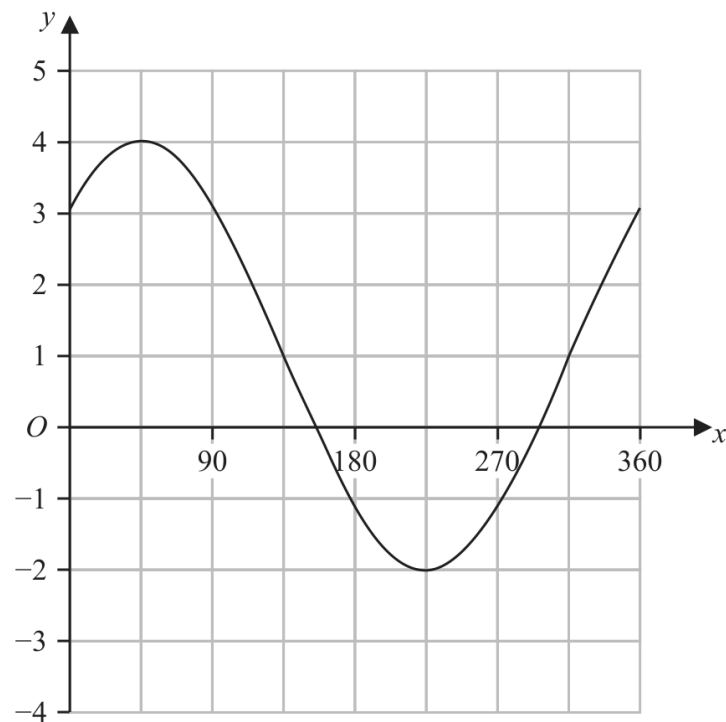
(Total for Question 25 is 2 marks)

Q45 **PAST PAPER** Edexcel 4MA1 · June 2025 · Paper 2HR · Q25

CH 3 · TOPIC 5 **3 marks**

PRACTISE · 3.5

25 The graph of $y = a \sin(x + b)^\circ + c$ for $0 \leq x \leq 360$ is drawn on the grid below.



Find a suitable value for a , for b and for c

$a = \dots\dots\dots$
 $b = \dots\dots\dots$
 $c = \dots\dots\dots$

(Total for Question 25 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------|---|
| 01 | Draw | Table, plot every point, one smooth curve. |
| 02 | Solve | Crossings with $y = 0$; other equations: draw the line (given – graph). |
| 03 | Shapes | Highest power and its sign; power on the bottom $\rightarrow$ never touches the axes. |
| 04 | Sketch | Complete the square $\rightarrow$ turning point; $x = 0$; $y = 0$. |
| 05 | Trig | Period 360° ; a height, b waves, c middle line. |

BOTTOM LINE Shape first $\rightarrow$ table and smooth curve $\rightarrow$ solve by reading crossings.

Answers

Graphs of Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • DRAW A CURVE FROM A TABLE

- | | | |
|------------|---|--|
| Q1 | Dr Eslam • Table | 5, 0, -3, -4, -3, 0, 5; minimum (1, -4) |
| Q2 | Practice • Easy | points joined with straight lines — it should be a smooth curve |
| Q3 | Practice • Easy | (a) 8, 4, 2, 1, 0.8, 0.5 (b) smooth curve |
| Q4 | Practice • Medium | (a) 2, 6 (b) U-curve (c) draw $y = x - 1$: $x \approx 1.6$, 4.4 |
| Q5 | Practice • Hard | (a) 3, 4.5, 1.5, 10.5 (b) curve (c) draw $y = -x - 1$: $x \approx -2.3$ |
| Q6 | Edexcel 4MA1 • January 2020 • Paper 2HR • Q15 | (a) 2, -1.5, -3, 0 (b) smooth U |
| Q7 | Edexcel 4MA1 • May 2021 • Paper 2H • Q15 | (a) 8.5, 5, 4, 5 (b) smooth curve, min (2, 4) |
| Q8 | Edexcel 4MA1 • January 2022 • Paper 1HR • Q4 | (a) 15, 8, 3, 0, -1, 0, 3 (b) U with minimum (2, -1) |
| Q9 | Edexcel 4MA1 • January 2022 • Paper 2HR • Q11 | (a) 12, 3, 1.5, 1.2 (b) curve |
| Q10 | Edexcel 4MA1 • June 2023 • Paper 2H • Q4 | (a) 8, -2, -4, -2, 2 (b) U-curve |
| Q11 | Edexcel 4MA1 • November 2024 • Paper 1H • Q11 | (a) 5, 4, 8.5, 10.4 (b) curve, min (1, 4) |
| Q12 | Edexcel 4MA1 • June 2025 • Paper 1H • Q10 | (a) -30, 15 (b) curve |

IDEA 02 • SOLVE EQUATIONS WITH A GRAPH

- Q13** Dr Eslam · Suitable line $x^2 - 3x = x - 1$: draw $y = x - 1$
- Q14** Practice · Easy (a) (1, 4) (b) $x \approx -0.4, 2.4$ (c) 3.75
- Q15** Practice · Easy $x \approx 1.7$
- Q16** Practice · Medium $k = 3$ or $k \approx -0.4$ (max and min)
- Q17** Practice · Hard (a) $x \approx -3, 1.5$ (b)(i) $a = 2, b = -5$ (ii) $x \approx -1.7, 1.2$
- Q18** Edexcel 4MA1 · January 2022 · Paper 2H · Q21 (a) $x \approx -0.2, 2.2$ (b) draw $y = 1 - 2x$: $x \approx -0.6, 1.6$
- Q19** Edexcel 4MA1 · November 2025 · Paper 2H · Q12 (a) 2.5, 2 (b) curve (c) draw $y = 3$: $x \approx 0.8, 5.2$

IDEA 03 · RECOGNISE THE SHAPE

- Q20** Practice • Easy (2.5, 0.4)
- Q21** Practice • Easy (i) straight line through O (ii) cubic through O
- Q22** Practice • Medium $(-1, 6)$
- Q23** Edexcel 4MA1 • January 2020 • Paper 2H • Q15 C, B, E
- Q24** Edexcel 4MA1 • June 2022 • Paper 2HR • Q16 C, F, D, H
- Q25** Edexcel 4MA1 • June 2023 • Paper 2HR • Q12 (a) D (b) E (c) B
- Q26** Edexcel 4MA1 • June 2023 • Paper 2H • Q18 C, F, A
- Q27** Edexcel 4MA1 • November 2023 • Paper 1H • Q12 (a) 3, 1, 3 (b) curve (c) D
- Q28** Edexcel 4MA1 • June 2024 • Paper 2HR • Q16 (i) E (ii) A
- Q29** Edexcel 4MA1 • June 2025 • Paper 2H • Q14 (a) B (b) C

IDEA 04 • SKETCH A QUADRATIC

- Q30** Dr Eslam · Sketch $(x - 3)^2 - 4$: TP $(3, -4)$; $(0, 5)$, $(1, 0)$, $(5, 0)$
- Q31** Practice · Easy $(1, -6)$
- Q32** Practice · Hard $A(-1, 3)$
- Q33** Practice · Hard (a) $A(-2, 9)$ (b) $y = 9 - 3(x - 2)^2$ (c) reflection in the x-axis
- Q34** Practice · Hard (a) TP $(2, -3)$, y-int $(0, 1)$ (b) $a = 4$, $b = 16$, $c = 12$
- Q35** Edexcel 4MA1 · January 2020 · Paper 1H · Q20 TP $(1, -a)$; $(0, 4 - a)$; $\left(1 \pm \frac{\sqrt{a}}{2}, 0\right)$
- Q36** Edexcel 4MA1 · November 2024 · Paper 1H · Q23 $(p - q)^2 = 64 + 4(9 + k) = 196$: $k = 24$

IDEA 05 · TRIGONOMETRIC GRAPHS

- Q37** Dr Eslam · Trig $(180, -1)$
- Q38** Practice · Easy (a) $P(180, 0)$, $Q(270, -1)$ (b) $\tan x$ through $0, 180, 360$
- Q39** Practice · Medium $\cos x$ from $(0, 1)$ to $(180, -1)$ to $(360, 1)$
- Q40** Practice · Medium (a) $P(180, 0)$ (b) $Q(270, -1)$
- Q41** Practice · Medium (a) $-k$ (b) k
- Q42** Edexcel 4MA1 · June 2022 · Paper 2H · Q23 (i) $A(180, 0)$ (ii) $B(360, -1)$
- Q43** Edexcel 4MA1 · June 2023 · Paper 2HR · Q26 $a = 2$, $b = 4$, $c = 1$
- Q44** Edexcel 4MA1 · June 2024 · Paper 2H · Q25 (i) $A(30, 2)$ (ii) $B(120, 0)$
- Q45** Edexcel 4MA1 · June 2025 · Paper 2HR · Q25 $a = 3$, $b = 45$, $c = 1$